COA - AT 1

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CSA0303 - Data Structure

1) Perform the following operations in LIFO data structure having an array size of 5. Push(45), Push(22), Push(11), Push(77), Pop(), Push(44), Push(33). print the peak element of data structure and its index position.

25/30

Ans

i) Push(45).

4	
3	
2	
1	
0	45

push = top+1
= -1+1
= 0.
Stack = [45]

ii) Push(22)

4	
3	
2	
1	22
0	45

push = top+1
= 0+1
= 1
stack = [22]

iii) push = top+1
= 1+1
= 2.
Stack = [11]

4	
3	
2	11
1	22
0	45

iv). Push (77)
= top+1
= 2+1
= 3
[stack] = 77

4	
3	77
2	11
1	22
0	45

v) Pop ().

Pop = top-1
= 3-1
= 2.
Stack = [11]

3	
2	11
1	22
0	45

vi) Push (44)
= top+1
= 2+1
= 3
stack = [44]

4	
3	44
2	11
1	22
0	45

vii) Push (33)

4	33
3	44
2	11
1	22
0	45

Push = top+1
= 3+1
= 4
[Stack] = 33

viii). Pop ()

4	
3	~~44~~
2	11
1	22
0	45

Pop = top - 1
= 4-1
= 3.
[Stack] = 44.

ix) Pop()

4	
3	
2	11
1	22
0	4

Pop = top-1
= 3-1
= 2.
[Stack] = 11

x) Pop()

4	
3	
2	
1	22
0	45

Pop = top-1
= 2-1
= 1
[Stack] = 22

(2) Perform the following operations in circular queue having a size of 5: Enqueue(20), Enqueue(10), En(30), Dq(), En(80), Dq(), Enq(90), Enq(100), Dq(), Dq(), Enq(60), Enq(90).

(i) Enq.

0	1	2	3	4
20				

Front = 0
Rear = rear+1
= -1+1
= 0

ii) En (10)

0	1	2	3	4
20	10			

Front = 0
rear = rear+1
= 0+1
= 1

iii) Eq (30).

20	10	30		
0	1	2	3	4

Front = 0

rear = rear + 1

= 1 + 1

= 2.

iv). Dq( ).

		10	30	
0	1	2	3	4

Fr = Frn + 1

= -1 + 1

= 0

v) Eq (80)

	10	30	80	
0	1	2	3	4.

Frnt = 0

= rear - 1

= 2 - 1

= 3

vi) Dq ( ).

		30	80	
0	1	2	3	4.

Frn = 1

rear = 0'

vii) Eq (90).

		30	80	90
0	1	2	3	4.

Frnt = 1

rear = rear + 1

= 3 + 1

= 4

vii). Eq(100).

100		30	80	90
0	1	2	3	4.

Front = 1

rear = -1 + 1

= 0

...vert the following infix expression into postfix expre...

B - (C/D +(E % F*G)/H) + J

Scan symbol	Stack	Postfix
B		B
-	-	B
(	- C	B
C	- C	BC.
/	- C /	BC
D	- C /	BCD
+	- C +	BCD/
(	- C + C	BCD/
E	- C + C	BCD/E
%	- C + C %	BCD/E
F	- C + C %	BCD/EF
*	- C + C *	BCD/EF%
G	- C + C *	BCD/EF%G
)	- C +	BCD/EF%G*+
)	-	BCD/EF%G*+
/	- /	BCD/EF%G*+H

H	-/	BCD/EF%G*+H/
*	-*	BCD/EF%G*+H/J
J	-*	BCD/EF%G*+H/J
End		BCD/EF%G*+H/J*-

BCD/EF%G*+H/J*-

9. 9384/*+

Symbol	Stack
9	9
3	93
8	938
4	9384
/	932
*	96
+	15

Answ - 15

10. 62+63/*

Symbol	Stack
6	6
2	62
+	8
6	86
3	863
/	82
*	16

Ans : 16

DQ

10			80	90
0	1	2	3	4

Fru = 1+1
= 2

Rear = 0

Not Completed.